Exponential functions

1 Would an exponential function generate the values shown in the following tables?

a

x	1	2	3	4	5	6
$f(x)$	5	25	125	625	3125	15 625

b

x	1	2	3	4	5	6
$f(x)$	4	9	11.5	15	13.5	11

2 What type of function would generate values as shown in the table?

x	1	2	3	4	5	6
$f(x)$	2	3	4.8	7.6	11.8	18

A Approximately exponential **B** Exactly exponential **C** Not exponential

3 Determine whether the following are increasing or decreasing exponential functions:

a $y = 10 \times \left(\frac{3}{5}\right)^x$ **b** $y = 9 \times 3^x$ **c** $y = 3 \times (0.5)^x$ **d** $y = 0.2 \times 2^x$

e $y = 2.2 \times 1.05^x$ **f** $y = 50 \times \left(\frac{7}{4}\right)^x$ **g** $y = 500 \times 0.75^x$ **h** $y = 80 \times 2^{-x}$

i $y = -5 \times 2^x$ **j** $y = -0.5 \times 0.8^x$ **k** $y = -2 \times \left(\frac{1}{4}\right)^x$ **l** $y = -4 \times 3^{-x}$

4 Which one of the following exponential functions rises most steeply?

A $y = 2 \times (2.1)^x$ **B** $y = 2 \times (2.2)^x$ **C** $y = 2 \times (1.7)^x$ **D** $y = 2 \times (1.2)^x$

5 For the following pairs of functions, select the function that increases more rapidly for $x > 0$:

a $y = 4^x$ and $y = 5^x$ **b** $y = 2^x$ and $y = 3 \times 2^x$

6 Find the missing coordinate in each ordered pair so that the pair is a solution of $y = 3^{-x}$:

a $(0, \square)$ **b** $(\square, 9)$ **c** $(-1, \square)$ **d** $\left(\square, \frac{1}{27}\right)$

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7 Find the missing coordinate in each ordered pair so that the pair is a solution of $y = -3^x$:

a $(5, \square)$ **b** $\left(\square, -\frac{1}{27}\right)$ **c** $(-1, \square)$ **d** $(\square, -81)$

8 Consider the function $f(x) = 3 + 4^{x+5}$. Evaluate $f(-2)$.

9 Consider the function $f(x) = \left(\frac{4}{3}\right)^x$.

a Evaluate $f(0)$. **b** Evaluate $f(-2)$.

10 Find the value of $5000(1.08^x)$ for $x = 30$ to two decimal places.

11 Find the value of $200\left(\frac{1.03^n - 1}{0.03}\right)$ for $n = 24$ to two decimal places.

12 Find the value of $P\left(1 + \frac{r}{k}\right)^{kn}$ for $P = 3000$, $r = 5\%$, $k = 4$ and $n = 2$ to two decimal places.

13 Consider the expression 2^x.

a Evaluate the expression when:

i $x = -4$ **ii** $x = 0$ **iii** $x = 4$

b For what values of x will 2^x have a negative value?

c Describe what happens to the value of 2^x as x gets larger.

d Describe what happens to the value of 2^x as x gets smaller.

14 Consider the function $y = 3^x$.

a Complete the table of values:

x	-5	-4	-3	-2	-1	0	1	2	3	4	5	10
y												

b Describe the behavior of the function as x increases.

c State the domain of the function.

d State the range of the function.

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15 Consider the function $y = 2^{-x}$.

a Complete the table of values:

x	-5	-4	-3	-2	-1	0	1	2	3	4	5	10
y												

b Describe the behavior of the function as x increases.

c Find the y-value of the y-intercept of the function.

d State the domain of the function.

e State the range of the function.

16 Consider the function $y = -(2^x)$.

a Complete the table of values:

x	-5	-4	-3	-2	-1	0	1	2	3	4	5
y											

b Can the value of y ever be zero or positive? Explain your answer.

c Is $y = -(2^x)$ an increasing or decreasing function?

d Describe the behavior of the function as x increases.

e State the domain of the function.

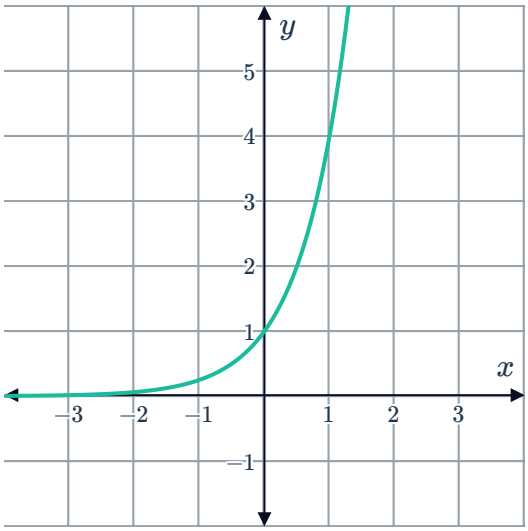
f State the range of the function.

17 Consider the graph of the equation $y = 4^x$:

a Is each y-value of the function positive or negative?

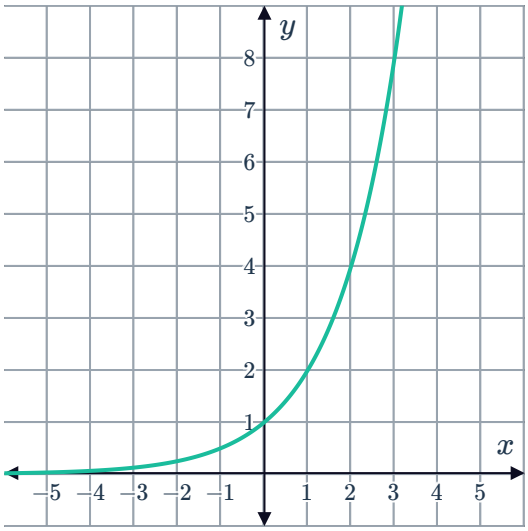
b State the value of y the graph approaches but does not reach.

c State the equation and name of the horizontal line, which $y = 4^x$ gets closer and closer to but never intersects.



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18 The graph of $y = 2^x$ is shown:



a State the y-intercept of this graph.

b Does the graph have an x-intercept?

c State the domain of the function.

d State the range of the function.

e Find the value of y when $x = 7$.

f Find the value of x when $y = 256$.

19 Consider the function $f(x) = e^x$.

a Complete the following table of values, correct to two decimal places:

x	-3	-2	-1	0	1	2	3
$f(x)$							

b Sketch the graph of $f(x) = e^x$.

20 Consider the function $y = -10^x$.

a Complete the following table:

x	-2	-1	0	1	2	3
y						

b For any value of x, is y always negative for this equation? Explain your answer.

c Sketch the graph of the function.

21 Consider the function $y = 8^{-x}$.

a Can the value of y ever be zero or negative? Explain your answer.

b As the value of x gets larger and larger, what value does y approach?

c As the value of x gets smaller and smaller, what value does y approach?

d Find the y-value of the y-intercept.

e How many x-intercepts does the curve have?

f Sketch the graph of $y = 8^{-x}$.

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22 Consider the function $y = 4\,(2^x)$.

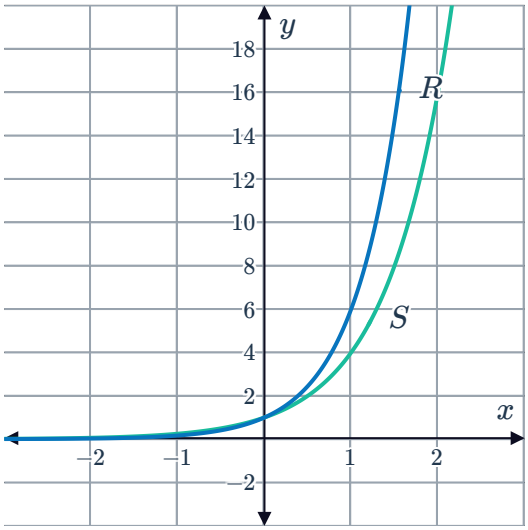
- **a** Find the y-value of the y-intercept.
- **b** Can the function value ever be negative?
- **c** As x approaches infinity, what value does y approach?
- **d** Sketch the graph of $y = 4\,(2^x)$.

23 Consider the function $y = 2\,(3^x)$.

- **a** Find the y-value of the y-intercept.
- **b** Can the function value ever be negative? Explain your answer.
- **c** State the domain.
- **d** State the range.
- **e** As x approaches positive infinity, what value does y approach?
- **f** Sketch the graph of $y = 2\,(3^x)$.

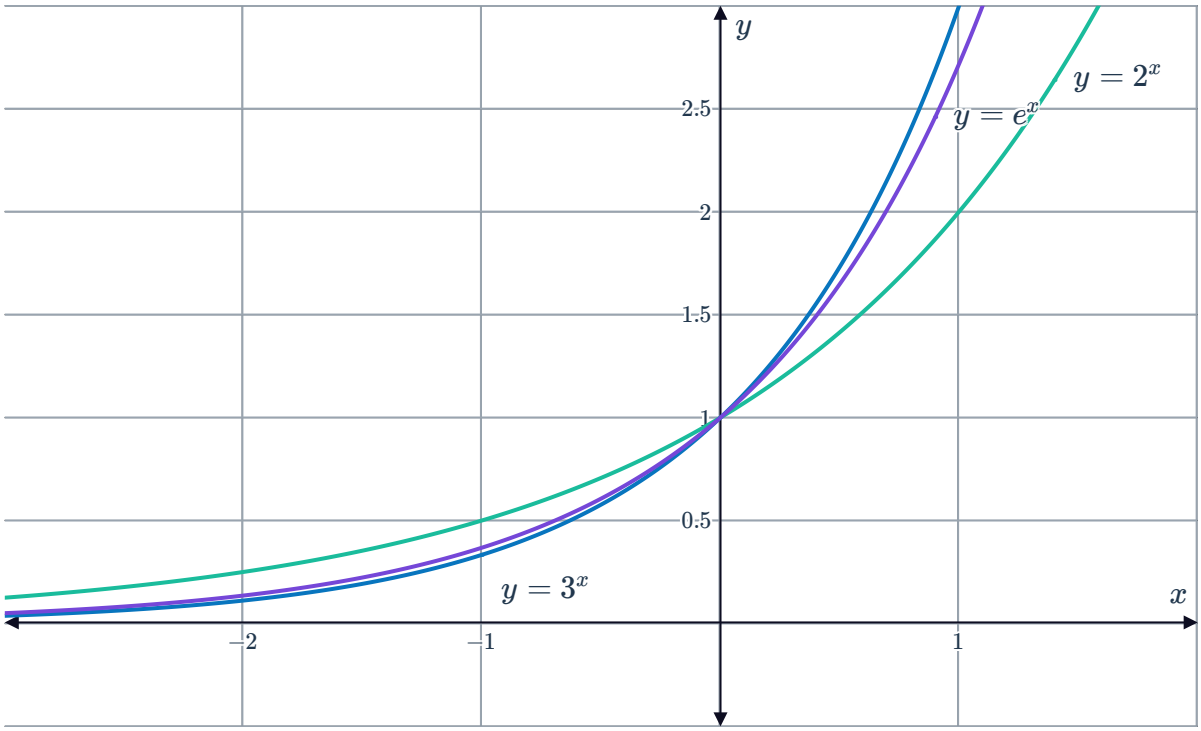
24 Consider the graphs of the two exponential functions R and S:

- **a** One of the graphs is of $y = 4^x$ and the other graph is of $y = 6^x$.
  Which is the graph of $y = 6^x$?
- **b** For $x < 0$, is the graph of $y = 6^x$ above or below the graph of $y = 4^x$. Explain your answer.



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25 The functions $y = 2^x$, $y = e^x$ and $y = 3^x$ have been sketched on the same coordinate plane:



a For what values of x does the inequality $3^x > e^x$ hold?

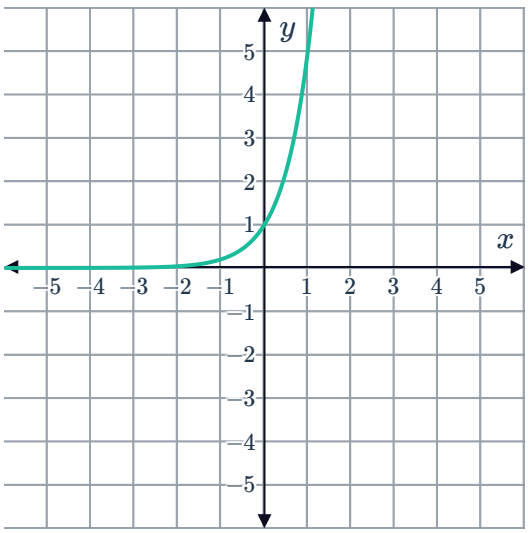
b For what values of x does the inequality $2^x > e^x$ hold?

Transformations of exponential functions

26 Consider the given graph of $y = 5^x$.

a Describe a transformation of the graph of $y = 5^x$ that would obtain $y = -5^x$.

b Sketch the graph of $y = -5^x$.

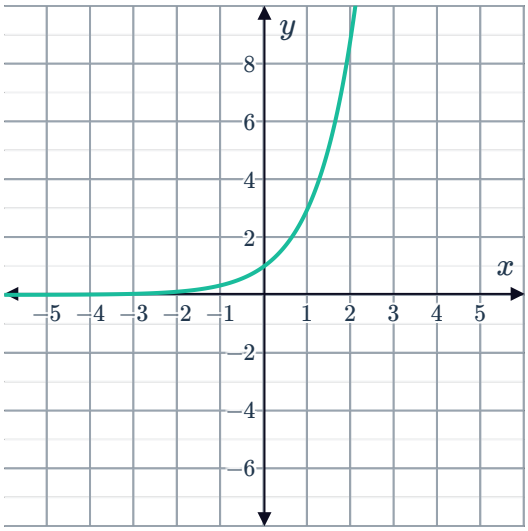


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27 Consider a graph of $y = 3^x$:

a Describe a transformation of the graph of $y = 3^x$ that would obtain the graph of $y = 3^x - 4$.

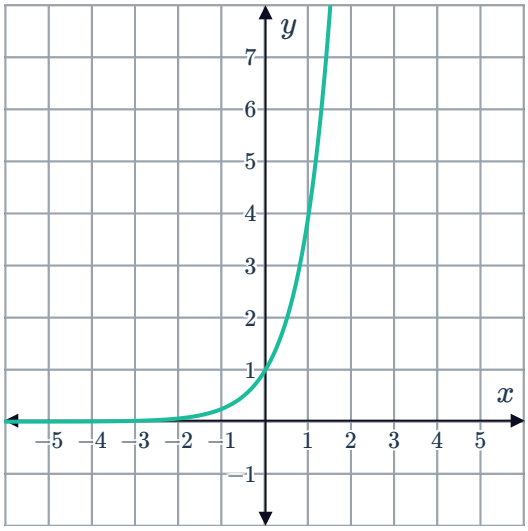
b Sketch the graph of $y = 3^x - 4$.



28 Consider the given graph of $y = 4^x$:

a Describe a transformation of the graph $y = 4^x$ that would obtain $y = 4^{-x}$.

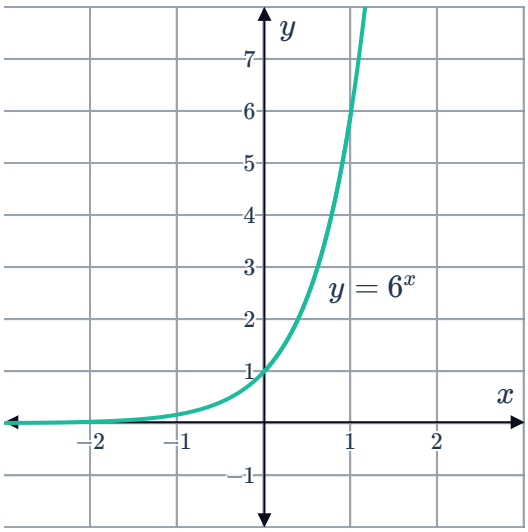
b Sketch the graph of $y = 4^{-x}$.



29 Consider the given graph of $y = 6^x$:

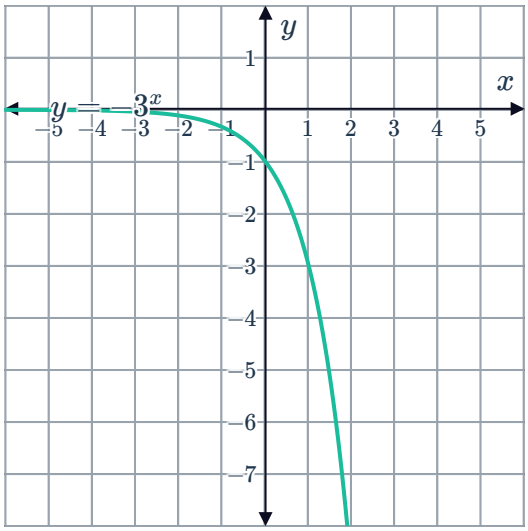
a Describe a transformation of the graph $y = 6^x$ that would obtain $y = -6^{-x}$.

b Sketch the graph of $y = -6^{-x}$.



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30 Consider the given graph of $y = -3^x$:



- **a** State the asymptote of $y = -3^x$.
- **b** Find the asymptote of $y = 2 - 3^x$.
- **c** How many x-intercepts would $y = 2 - 3^x$ have?
- **d** State the domain of $y = 2 - 3^x$.
- **e** State the range of $y = 2 - 3^x$.

31 Consider the functions $y = 2^x$ and $y = 2^x - 2$.

- **a** Find the y-intercept of $y = 2^x$.
- **b** Determine the y-intercept of $y = 2^x - 2$.
- **c** State the horizontal asymptote of $y = 2^x$.
- **d** Determine the horizontal asymptote of $y = 2^x - 2$.

32 For each of the following functions, determine the following:

- **i** Horizontal translation
- **ii** Vertical translation
- **iii** Growth constant

a $y = 5^{x+4} + 2$ **b** $y = 5^{3x+3} - 1$ **c** $y = 2^{5x-1} + 2$

33 Find the equation of the new function that results from the following transformations:

- **a** The graph of $y = 2^x$ is translated down by 7 units.
- **b** The graph of $y = 11^x$ is moved to the left by 10 units.

34 For each of the following, write the equation of the new function that results from $y = 7^x$ after the following translations:

- **a** Translated three units to the right and then five units downward.
- **b** A reflection in the x-axis, then translated five units upward and three units to the left.
- **c** Translated five units upward, three units to the left and then reflected in the x-axis.

35 Sketch the graph of the following functions:

a $y = 2^{x+4}$ **b** $y = 2^x + 4$

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36 For each of the following exponential functions:

i Find the y-value of the y-intercept.

ii State the domain.

iii State the range.

iv Sketch the graph of the function.

a $y = 4^x + 3$ **b** $y = 2^x - 5$ **c** $y = 2^{x+4}$ **d** $y = 3^{x-1}$

e $y = 3 - 2^{-x}$

37 Consider the function $y = -5^x$.

a State the equation of the asymptote of $y = -5^x$.

b State the equation of the asymptote of $y = 2 - 5^x$.

c How many x-intercepts would the graph of $y = 2 - 5^x$ have?

38 For each of the following functions:

i Find the y-intercept of the curve.

ii State the equation of the horizontal asymptote.

iii Sketch the graph of the function.

a $y = 3^x + 2$ **b** $y = 2^x - 2$ **c** $y = -3^x + 2$ **d** $y = 3^{-x} - 1$

e $y = 3^{-x} + 1$

39 Consider the original graph $y = 3^x$. The function values of the graph are multiplied by 2 to form a new graph.

a For each point on the original graph, find the point on the new graph:

Point on original graph	Point on new graph
$\left(-1, \frac{1}{3}\right)$	$(-1, \square)$
$(0, 1)$	$(0, \square)$
$(1, 3)$	$(1, \square)$
$(2, 9)$	$(2, \square)$

b State the equation of the new graph.

c Graph the original and new graph on the same coordinate plane.

d Describe the postion of new graph in relation to the original graph.

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40 Using a graphing calculator, graph each set of three functions below on the same screen and state whether they have the same:

i y-intercept ii Asymptote iii Range

a $y = e^x$, $y = e^x + 2$, and $y = e^x - 3$

b $y = e^x$, $y = e^{x-2}$, and $y = e^{x+3}$

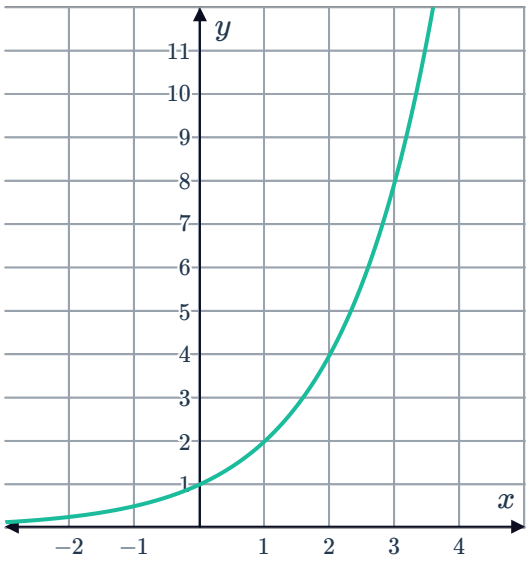
c $y = e^x$, $y = e^{2x}$, and $y = e^{3x}$

d $y = e^x$, $y = 4e^x$, and $y = -4e^x$

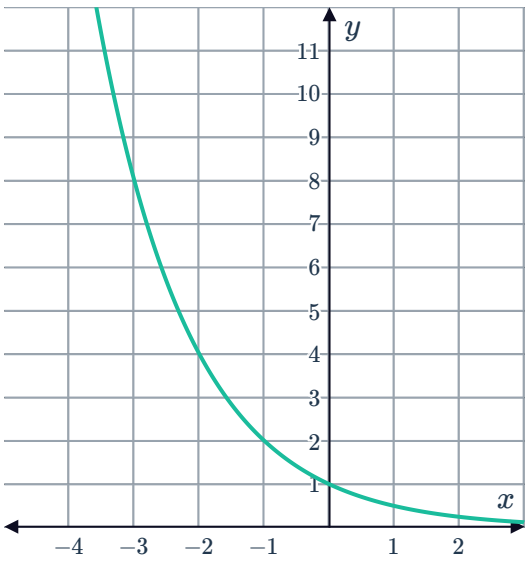
Exponential equations

41 Find the equation of the following graphs:

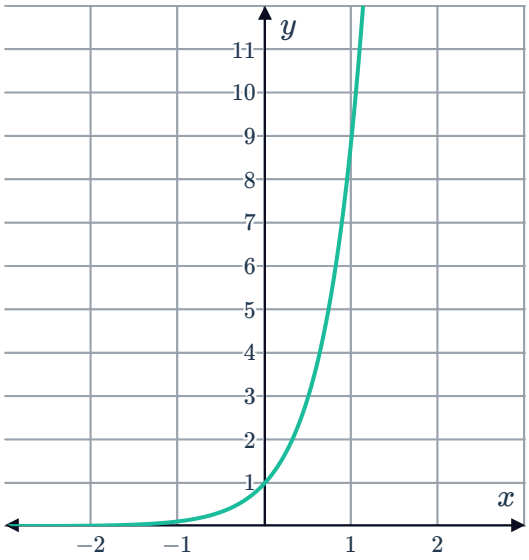
a



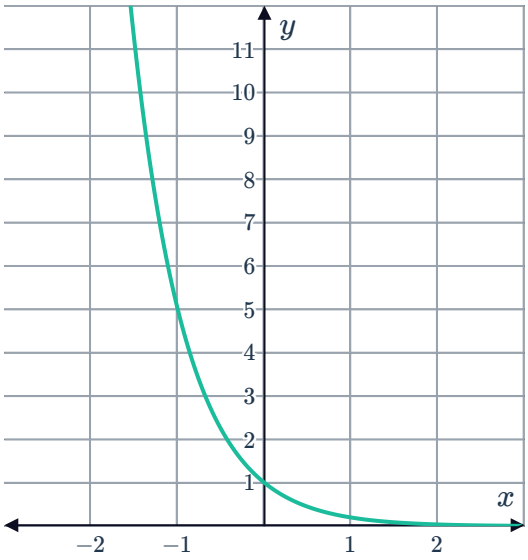
b



c

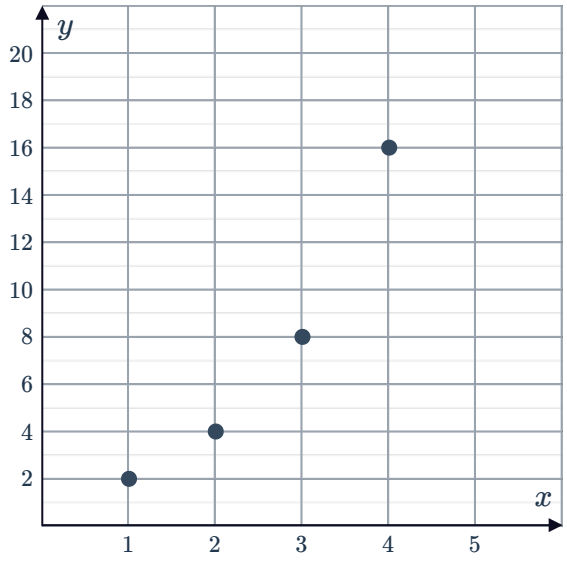


d



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42 Consider the points given in the graph:



a Complete the table of values:

x	1	2	3	4
y				

b Identify the common ratio between consecutive y values.

c State the equation relating x and y.

43 Consider the given table of values:

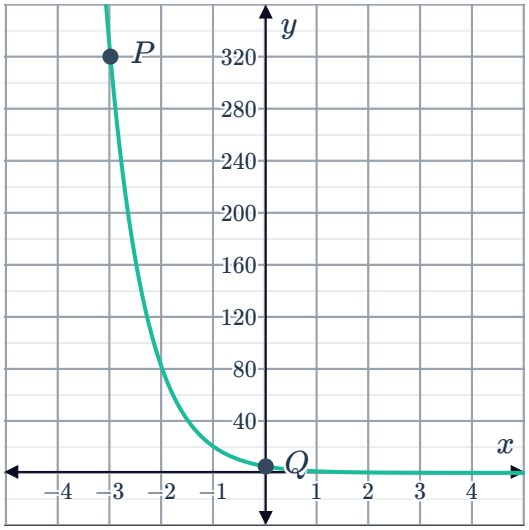
x	1	2	3	4
y	10	100	1000	10 000

a Identify the common ratio between consecutive y values.

b State the equation relating x and y.

c Find the value of y when $x = 10$.

44 The points $P\,(-3, 320)$ and $Q\,(0, 5)$ satisfy an exponential function of the form $y = ab^{-x}$.



a Find the value of a.

b Find the value of b.

c Find the value of y when $x = 2$.

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